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Semester Project

Health Inequality In Andhra Pradesh and Jammu & Kashmir

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1 Introduction

India's states differ substantially in economic and health services. Two states, Andhra Pradesh(Southern), with a stronger rural-urban infrastructure, contrasting with Jammu and Kashmir(Northern), a relatively less developed state. This study aims to provide a comparative analysis of Andhra Pradesh and Jammu and Kashmir on two dimensions: (1) Household consumer expenditure inequality across subgroups(like gender, etc.); and (2) Health quality at district level. Whenever relevant, statistical tests are used to evaluate whether the differences between the two states are significant or not.

2 Socio-Economic Profile of Andhra Pradesh and Jammu& Kashmir

This section provides a brief overview of their key socio-economic indicators.

2.1 Economic Indicator

Andhra Pradesh is one of the India's relatively prosperous states. Recent data show that Andhra Pradesh's Gross State Domestic Product(GSDP) is much higher than GSDP of Jammu and Kashmir.
Also we can see that the GSDP of Andhra Pradesh has increased sharply across time whereas increase of GSDP in Jammu And Kashmir more stable, thus widening the gap.

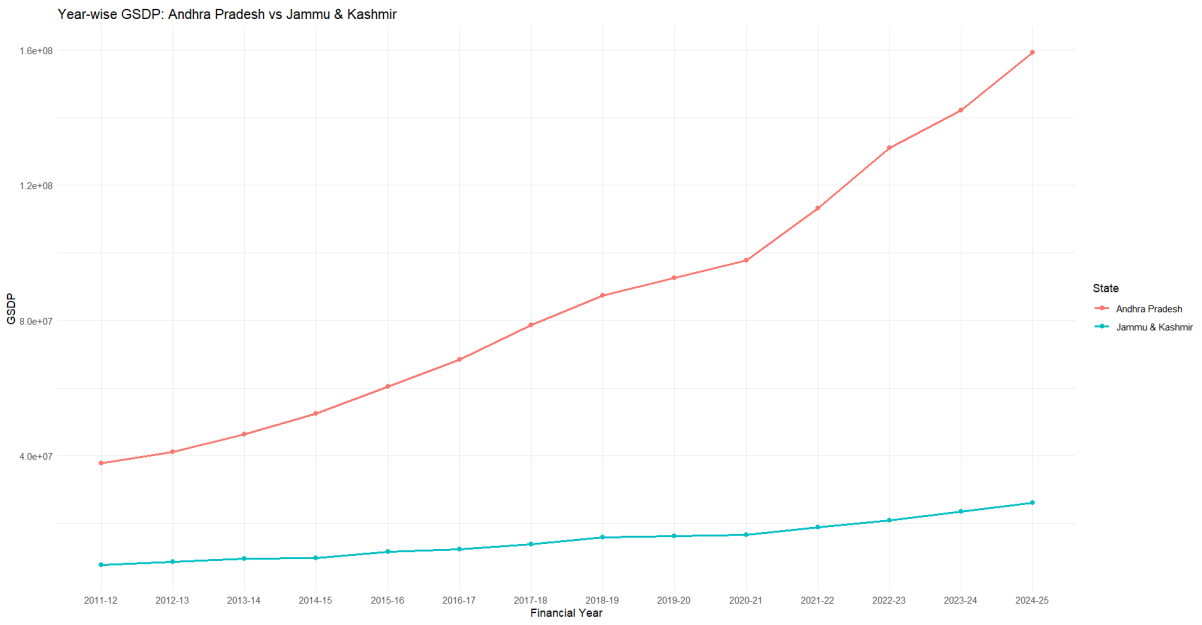


Fig.2.1 Year wise GSDP plot (Using RBI data), Red - AP, Blue-J&K

2.2 Descriptive Statistics And Visualisation

Following population pyramids (age-sex pyramid) depict the proportion of various age groups for male and females in Andhra Pradesh and Jammu & Kashmir based on the household health-survey dataset for India, from the National Sample Survey (NSS), 80th Round.

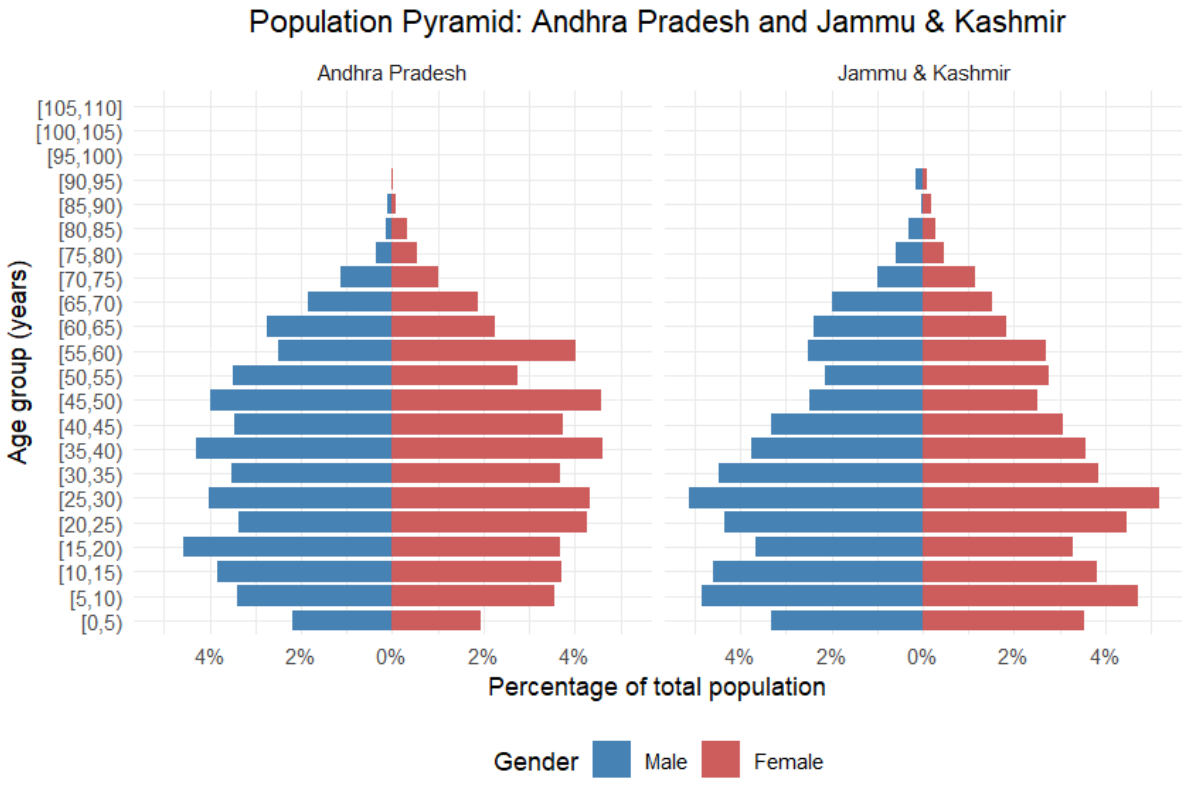


Fig.2.2 Population Pyramid (using final weights)

The pyramids indicate that Jammu and Kashmir has relatively younger population, while Andhra Pradesh shows a more balanced structure with comparitively smaller base. This is also reflected in the dependency ratio below.

- Andhra Pradesh
  - Total Dependency Ratio: 35.3 %
  - Child Dependency Ratio: 25.2 %
  - Old-Age Dependency Ratio: 10.1 %
- Jammu and Kashmir
  - Total Dependency Ratio: 48.2 %
  - Child Dependency Ratio: 36.8 %
  - Old-Age Deoendency Ratio: 11.4 %

We can observe that Jammu and Kashmir has higher dependency ratios than Andhra Pradesh. These findings suggest that Jammu & Kashmir has a greater relative demographic burden associated with younger dependents, potentially implying greater demand for education and child-related services.

2.3 Interpretation

These basic comparisons show that Andhra Pradesh and Jammu and Kashmir have different development trajectories, making them an ideal pair for exploring inequality in health-related factors.

3 Research Question And Data Description

Having stated the socio-ecomic condition of Andhra Pradesh and Jammu & Kashmir, we will now look at the research question that guide this study. the analysis is based on four major component , Household usual consumer expenditure (umce in data), Chronic Ailments, Recent Ailment, Hospitalisation and other health related variables.

3.1 Research Question

The study aims to answer the following question:

1. How does umce inequality compare between Andhra Pradesh and Jammu and Kashmir?
2. Within each state how does the health quality indicators differ across subgroups such as gender?
3. Do State and socio economic backgroud(umce) have significant effect in the health quality indicators?
4. Do State and gender have significant effect on health quality Indicators?

Through tthese questions we aim to capture the inequalaity in Health Quality in Andhra Pradesh and Jammi & Kashmir.

3.2 Datasets Used

We have used NSSO 80th round dataset on health survey. Following variables from the datasets were used.

- Household Usual Consumption Expenditure(umce): The umce provides detailed information on household-level expenditure across both rural and urban sectors.

- Age
- Gender
- final_weights
- whether hospitalised/ No. of times hospitalised
- Whether suffered from any chronic ailment (b3c14)

4 Methodology

4.1 The Lorenz Curve

The Lorenz curve, introduced by Max O. Lorenz (1905), is a fundamental graphical device for representing the distribution of a nonnegative resource (typically expenditure or income) across a population. Ordering individuals from poorest to richest, the Lorenz curve plots the cumulative share of total resources received by the bottom p fraction of the population against $p \in [0, 1]$. The curve provides an immediate visual sense of inequality: "closeness" to the diagonal line $L(p) = p$ indicates greater equality, while larger deviations indicate greater concentration.

Mathematical Definition

Let X be a nonnegative random variable representing the expenditure. We impose the following conditions on the distribution F of X:

1. **Nonnegativity:** $P(X \geq 0) = 1$.
2. **Finite positive mean:** $\mu := E[X] \in (0, \infty)$.
3. **Absolute continuity (regularity):** F admits a density f on $(0, \infty)$ and $f(x) > 0$ on the interior of its support.
4. **Strictly increasing CDF on support:** F is strictly increasing on $(0, \infty)$, so that the (generalized) inverse $F^{-1}(p)$ is well-defined and differentiable almost everywhere on $(0, 1)$.

Define two auxiliary functions that summarize population shares and expenditure-mass shares up to cutoff x:

$$F_0(x) := P(X \leq x) = F(x) \qquad \text{(population CDF)},$$
$$F_1(x) := \frac{1}{\mu} E\left[X \mathbf{1}_{\{X \leq x\}}\right] = \frac{1}{\mu} \int_0^x t \, dF(t) \qquad \text{(cumulative expenditure share up to } x\text{)}.$$

The Lorenz curve $L : [0,1] \to [0,1]$ is the composition

$$L(p) := F_1\left(F_0^{-1}(p)\right) = \frac{1}{\mu}\int_0^{F_0^{-1}(p)} t\, dF(t), \quad 0 \le p \le 1,$$

where the generalized inverse is given by $F_0^{-1}(p) := \inf\{x : F_0(x) \ge p\}$. Equivalently, $L(p)$ is the fraction of total expenditure spent by the bottom p fraction of the population.

- $L(0) = 0$ and $L(1) = 1$ (all mass accounted for at the endpoints).
- L is nondecreasing and continuous from the right.
- Under the differentiability assumptions above, for almost every $p \in (0,1)$,

$$L'(p) = \frac{F_0^{-1}(p)}{\mu},$$

  i.e., the slope of L equals the corresponding quantile normalized by the mean.
- Since $F_0^{-1}(p)$ is nondecreasing in p, $L'(p)$ is nondecreasing and therefore L is convex.

Discrete (Empirical) Setup and Computation

In applications, one commonly works with a finite sample of expenditure values $\{x_i\}_{i=1}^n$. Let $x_{(1)} \le x_{(2)} \le \cdots \le x_{(n)}$ denote the sorted expenditure values, and set total expenditure $T = \sum_{i=1}^n x_{(i)}$ and sample mean $\bar{x} = T/n$.

Define empirical step functions:

$$\hat{F}_0(x) = \frac{1}{n}\sum_{i=1}^n \mathbf{1}_{\{x_{(i)} \le x\}}, \quad \hat{F}_1(x) = \frac{1}{T}\sum_{i=1}^n x_{(i)}\mathbf{1}_{\{x_{(i)} \le x\}}.$$

The empirical Lorenz curve $\hat{L}(p)$ is the piecewise-linear curve obtained by joining the points

$$\left(p_i, \hat{L}(p_i)\right) = \left(\frac{i}{n}, \frac{\sum_{j=1}^i x_{(j)}}{T}\right), \quad i = 0, 1, \ldots, n,$$

(where $\sum_{j=1}^0 \cdot = 0$ and $p_0 = 0, p_n = 1$). This construction is the empirical analogue of $L(p)$ and converges to L under mild regularity as $n \to \infty$.

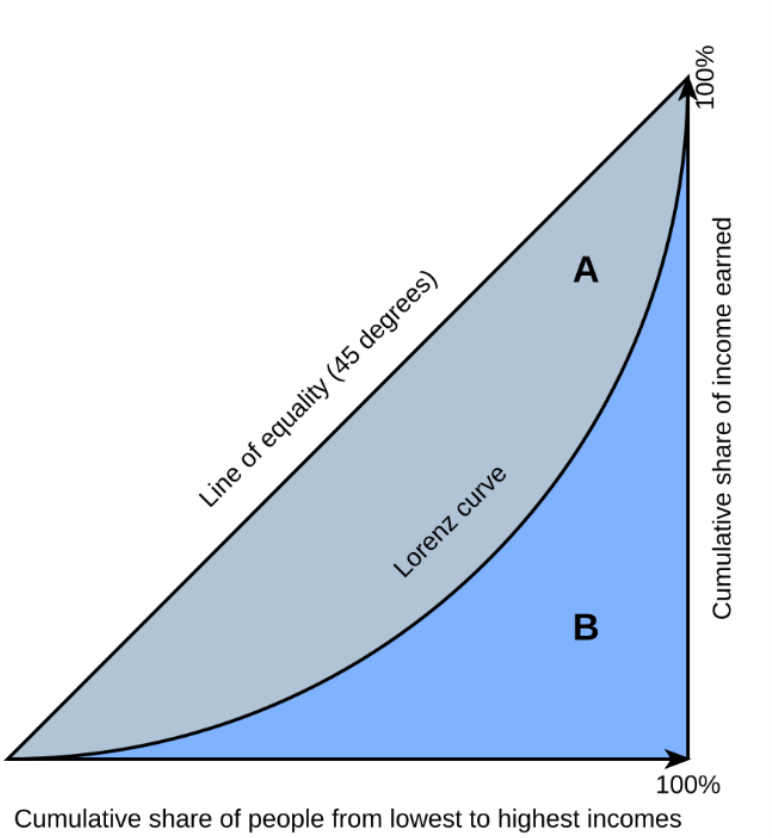


Fig.4.1 Lorenz Curve

4.1.1 Weighted Lorenz Curve

In survey data, observations may represent different numbers of individuals in the target population. Let $w_i > 0$ denote the survey weight associated with observation i, and let $x_i \geq 0$ denote its per capita expenditure.

Arrange the observations in ascending order of expenditure, $x_{(1)} \leq \cdots \leq x_{(n)}$, and let $w_{(i)}$ denote the weight corresponding to $x_{(i)}$. Define the total survey weight and weighted total expenditure as

$$W = \sum_{i=1}^{n} w_{(i)}, \qquad T_w = \sum_{i=1}^{n} w_{(i)} x_{(i)}.$$

The cumulative weighted population share is

$$P_i^{(w)} = \frac{\sum_{j=1}^{i} w_{(j)}}{W},$$

and the cumulative weighted expenditure share is

$$L_i^{(w)} = \frac{\sum_{j=1}^{i} w_{(j)} x_{(j)}}{T_w}.$$

The weighted Lorenz curve is obtained by joining the points

$$\left(\frac{\sum_{j=1}^{i} w_{(j)}}{W}, \frac{\sum_{j=1}^{i} w_{(j)} x_{(j)}}{T_w} \right), \quad i = 0, 1, \ldots, n.$$

At $i = 0$, both cumulative shares are zero. At $i = n$, both shares equal one. The weighted Lorenz curve accounts for differences in the population represented by each observation.

4.2 Gini Index

The Gini Index, introduced by Corrado Gini in 1912, is one of the most widely used measures of inequality. It quantifies how much the actual distribution of expenditure or income deviates from a state of perfect equality (Egalitarian scenario). The Gini Index represents the ratio of the area between the Lorenz curve and the line of perfect equality to the total area under the line of perfect equality. A Gini value of 0 indicates complete equality (everyone has the same expenditure value), while a value of 1 reflects extreme inequality, where one individual possesses all the expenditure.

Relation to the Lorenz Curve

As the Gini coefficient is the ratio of the area between the line of equality and the Lorenz curve to the total area under the line of equality, denote by A the area between the line of equality and the Lorenz curve, and by B the area under the Lorenz curve. Since the total area under the line of equality equals $1/2$, we can write:

$$G = \frac{A}{A + B} = 2A = 1 - 2B.$$

$$G = 1 - 2 \int_0^1 L(p)\, dp = 2 \int_0^1 \big(p - L(p)\big)\, dp.$$

Under absolute continuity, change variables $p = F_0(x)$ and use Fubini's theorem to obtain alternative representations, e.g.,

$$G = 1 - \frac{2}{\mu} \int_0^\infty x\big(1 - F(x)\big)\, dF(x) = \frac{E[|X - X'|]}{2\mu},$$

where X, X' are independent draws from F.

Discrete Gini Formulas

Approximating the integral $\int_0^1 L(p)\,dp$ by the trapezoidal rule yields the sample Gini:

$$\hat{G} = 1 - 2\sum_{i=1}^{n} \hat{L}\left(\frac{i}{n}\right)\frac{1}{n} = 1 - \frac{2}{n}\sum_{i=1}^{n}\frac{\sum_{j=1}^{i} x_{(j)}}{T} = \frac{1}{2n^2\bar{x}}\sum_{i=1}^{n}\sum_{j=1}^{n}|x_i - x_j|.$$

This formula calculates the average absolute difference between all pairs of individuals, normalized by the mean expenditure.

An equivalent and computationally convenient expression (for sorted data $x_{(1)} \leq \cdots \leq x_{(n)}$) is:

$$G = \frac{2}{n^2\bar{x}}\sum_{i=1}^{n} i x_{(i)} - \frac{n+1}{n}.$$

4.2.1 Weighted Gini Index

The weighted Gini coefficient accounts for differences in survey weights and is defined using the weighted Lorenz curve as

$$\widehat{G}_w = 1 - 2\int_0^1 \widehat{L}_w(p)\,dp.$$

Using the trapezoidal rule, it can be computed as

$$\widehat{G}_w = 1 - \sum_{i=1}^{n}\left(L_{i-1}^{(w)} + L_i^{(w)}\right)\left(P_i^{(w)} - P_{i-1}^{(w)}\right),$$

where $P_0^{(w)} = L_0^{(w)} = 0$.

An equivalent pairwise-difference expression is

$$\widehat{G}_w = \frac{\sum_{i=1}^{n}\sum_{j=1}^{n} w_i w_j |x_i - x_j|}{2W\sum_{i=1}^{n} w_i x_i}.$$

Defining the weighted mean expenditure as

$$\bar{x}_w = \frac{\sum_{i=1}^{n} w_i x_i}{W},$$

the weighted Gini coefficient can also be expressed as

$$\widehat{G}_w = \frac{\sum_{i=1}^{n}\sum_{j=1}^{n} w_i w_j |x_i - x_j|}{2W^2 \bar{x}_w}.$$

The weighted Gini coefficient ranges from zero to one for nonnegative expenditure with positive weighted mean. A value of zero indicates perfect equality, whereas larger values indicate greater expenditure inequality.

4.3 Logistic Regression

4.3.1 Define the Response Variable

Let Y_{ijk} denote the hospitalisation status of individual k in state i and gender category j:

$$Y_{ijk} = \begin{cases} 1, & \text{if hospitalised,} \\ 0, & \text{otherwise.} \end{cases}$$

Let $p_{ijk} = P(Y_{ijk} = 1)$.

4.3.2 Fit a Logistic Regression Model

$$\log\left(\frac{p_{ijk}}{1-p_{ijk}}\right) = \beta_0 + \alpha_i + \beta_j + (\alpha\beta)_{ij}.$$

Here, α_i represents the state effect, β_j represents the gender effect, and $(\alpha\beta)_{ij}$ represents their interaction.

4.3.3 Test the Three Effects

Effect	**Null hypothesis**
State	No state effect, conditional on gender
Gender	No gender effect, conditional on state
State × Gender	No interaction

For each effect, use a statistical test to obtain a p-value. At significance level 0.05, reject the corresponding null hypothesis if $p < 0.05$.

With an interaction in the model, the main effects depend on the reference categories. For a complete assessment, test the interaction jointly and compare the state and gender effects using appropriate contrasts or predicted probabilities.

5 Results And Conclusions

5.1 Inequality Analysis

Table.5.1 Gini Index

Statistic	Andhra Pradesh	Jammu & Kashmir
Gini	0.304	0.250

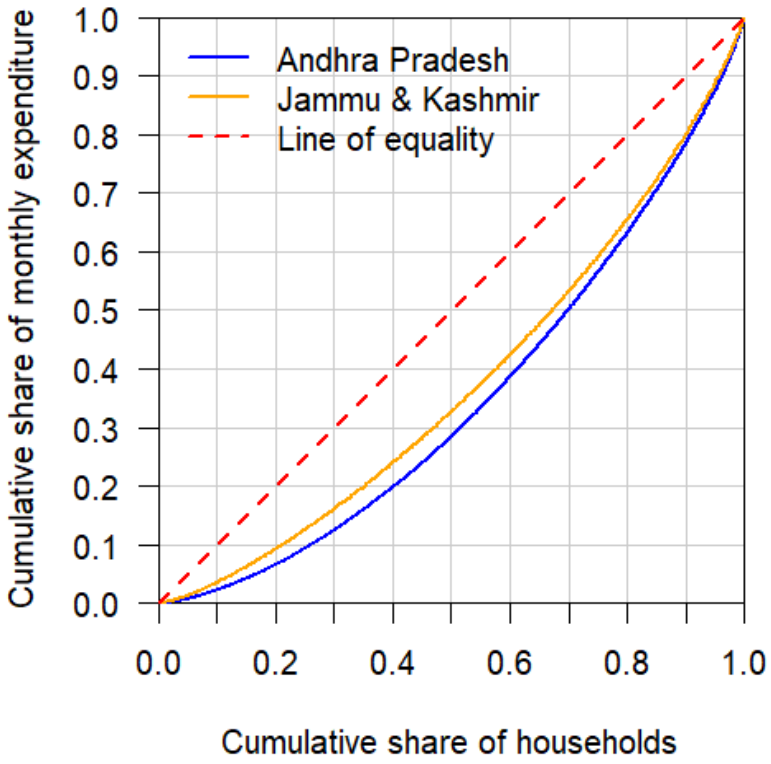


Fig.5.1 Lorenz Curves of Household Expenditure

5.2 Hospitalisation Prevalance Trend In Male and Female Across Different Expenditure Groups

$$\text{Prevalence (\%)} = \frac{\text{Number of individuals with the condition}}{\text{Total number of individuals in the subgroup}} \times 100$$

The Fig.5.2 graph shows how the Hospitalisation varies in Male and Female with increase in expenditure(Here we assume that people with higher expenditure are richer).

The graph shows that female hospitalisation prevalence is higher than male prevalence across all expenditure groups in both Andhra Pradesh and Jammu & Kashmir. Hospitalisation prevalence generally increases with expenditure for both genders. In Andhra Pradesh, the gender gap narrows considerably in the highest expenditure group, whereas in Jammu & Kashmir, it remains relatively stable. These are unweighted descriptive

findings and may reflect differences in age, health status, healthcare access and other factors; they do not establish causation.

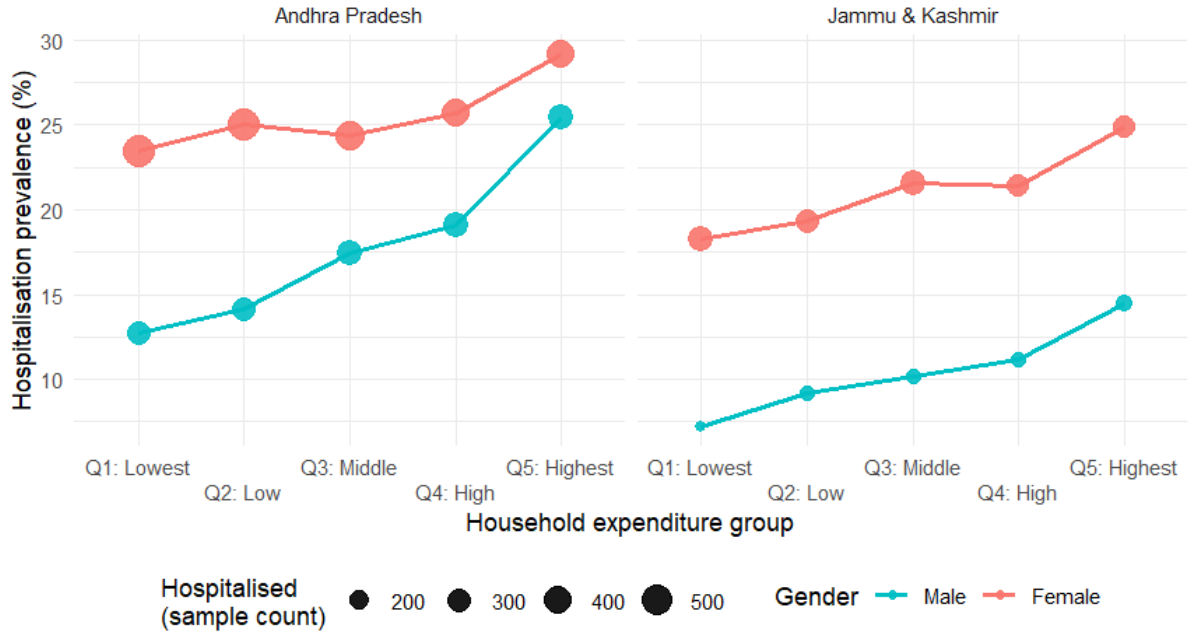


Fig.5.2 Hospitalisation Prevalence across expenditure groups

The population pyramid shows that both Andhra Pradesh and Jammu & Kashmir have relatively young populations, with a substantial share of individuals in younger and working-age groups. However, their age distributions differ: Jammu & Kashmir has a particularly prominent 25–30 age group, while Andhra Pradesh has relatively large

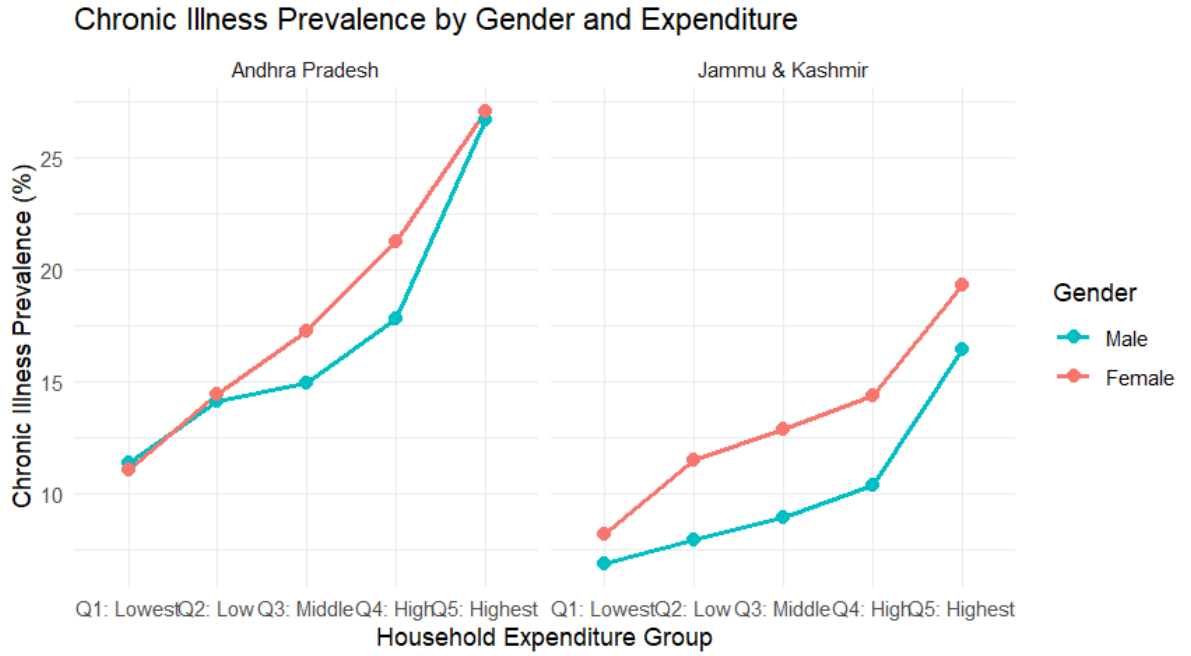


Fig.5.3 Chronic ailment across expenditure groups

groups across several working-age ranges, including 35–50 years. We can give this as a explanation to prevalance of Andhra Pradesh being higher than Jammu & Kashmir.

In Fig. 5.3 we can see that chronic ailment is increasing with in expenditure (group) this gives an explation to the previous graph Fig.5.2 because people with chronic ailmnets are very likely to visit the Hospitals.

5.3 Logistic Regression Analysis of Hospitalisation Based on Gender

Objective

The objective of this analysis is to examine the association between state, gender, monthly consumer expenditure, and age with the probability of hospitalisation among individuals in Andhra Pradesh and Jammu & Kashmir. Particular attention is given to whether the association between gender and hospitalisation differs across the two states.

The analysis includes all individuals with valid observations on the response and explanatory variables, irrespective of their chronic illness status. The model is estimated using unweighted observations.

Note that Gender 1 is Male, Gender 2 is Female, and Gender 3 is Transgender.

Model Specification

The logistic regression model incorporating the interaction between state and gender is specified as

$$\begin{aligned}\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 S_i + \beta_2 G_{2i} + \beta_3 G_{3i} \\ + \beta_4 U_i + \beta_5 A_i \\ + \beta_6 (S_i G_{2i}) + \beta_7 (S_i G_{3i}),\end{aligned}$$

where

- S_i is an indicator variable equal to 1 for Jammu & Kashmir and 0 for Andhra Pradesh.
- G_{2i} is an indicator variable equal to 1 for Gender 2 and 0 otherwise.
- G_{3i} is an indicator variable equal to 1 for Gender 3 and 0 otherwise.
- U_i denotes monthly consumer expenditure.
- A_i denotes the age of the individual in years.
- $\beta_0, \ldots, \beta_7$ are the regression coefficients.

Variable	Estimate	Std. Error	z-value	p-value	Odds Ratio
Intercept	-1.9760	0.0504	-39.197	< 0.001	0.139
State: Jammu & Kashmir	-0.5581	0.0491	-11.363	< 0.001	0.572
Gender 2	0.5105	0.0376	13.569	< 0.001	1.666
Gender 3	1.2300	0.6905	1.781	0.0749	3.421
umce	-1.534×10^{-5}	1.735×10^{-6}	-8.846	< 0.001	0.999985
Age	0.01871	0.0006981	26.808	< 0.001	1.019
State $\times$ Gender 2	0.3586	0.06236	5.750	< 0.001	1.431
State $\times$ Gender 3	-1.9630	1.2460	-1.576	0.1151	0.140

Note: The response variable is hospitalisation status (1 = hospitalised, 0 = not hospitalised). Andhra Pradesh and Gender 1 are the reference categories. The model adjusts for age and monthly consumer expenditure. Odds ratios are calculated as $\exp(\hat{\beta})$. For interaction terms, the odds ratios represent ratios of odds ratios. The analysis uses unweighted observations.

Andhra Pradesh and Gender 1 are the reference categories. The interaction terms allow the association between gender and hospitalisation to differ between the two states.

Hypotheses for the Interaction Effect

The null hypothesis states that the association between gender and hospitalisation does not differ across the two states:

$$H_0 : \beta_6 = \beta_7 = 0.$$

The alternative hypothesis states that at least one interaction coefficient is non-zero:

$$H_1 : \text{at least one of } \beta_6, \beta_7 \text{ is non-zero.}$$

These hypotheses are examined using a likelihood-ratio test comparing the main-effects model with the interaction model.

Likelihood-Ratio Test

The main-effects model includes state, gender, monthly consumer expenditure, and age, without interaction terms. The interaction model additionally includes the state–gender interaction terms.

The likelihood-ratio statistic is calculated as

$$G^2 = D_{\text{main}} - D_{\text{interaction}},$$

where D_{main} and $D_{\text{interaction}}$ denote the residual deviances of the two models.

The observed results are

$$D_{\text{main}} = 29572,$$
$$D_{\text{interaction}} = 29535,$$
$$G^2 = 37.115.$$

The difference in the degrees of freedom is 2. Therefore, the test statistic is compared with a chi-squared distribution with two degrees of freedom.

$$\chi^2(2) = 37.115, \qquad p = 8.723 \times 10^{-9}.$$

Since the p-value is less than 0.05, the null hypothesis is rejected. The interaction model provides a significantly better fit than the main-effects model. This indicates that the association between gender and hospitalisation differs between Andhra Pradesh and Jammu & Kashmir.

The regression results indicate that, for Gender 1, the odds of hospitalisation in Jammu & Kashmir are 42.8% lower than in Andhra Pradesh (OR = 0.572, $p < 0.001$). The interaction between state and Gender 2 is statistically significant (OR = 1.431, $p < 0.001$), indicating that the gender association with hospitalisation differs between the states. However, the interaction involving Gender 3 is not significant (p = 0.1151). The likelihood-ratio test confirms that including the state–gender interaction significantly improves model fit, $\chi^2(2) = 37.115,\ p < 0.001$. These findings represent associations from an unweighted model and do not establish causality.

In Andhra Pradesh, Gender 2 has 66.6% higher odds of hospitalisation than Gender 1 (OR = 1.666, $p < 0.001$), whereas the difference for Gender 3 is not statistically significant (p = 0.0749). An increase of 1,000 expenditure units is associated with approximately 1.52% lower odds of hospitalisation, while each additional year of age is associated with 1.89% higher odds; both associations are statistically significant ($p < 0.001$).

6 Policy And Practical Implications

6.1 For Andhra Pradesh

- Target healthcare support towards lower-expenditure groups: Our unweighted hospitalisation prevalence declines from 28.2% in the lowest expenditure quintile to 18.2% in the highest quintile in Andhra Pradesh. This pattern warrants investigating whether lower-expenditure households face unmet healthcare needs or barriers to hospitalisation. Policy could prioritise access assessments and referral support for poorer households rather than assuming that lower hospitalisation means better health.

- Gender differences in hospitalisation. Your logistic regression estimates higher odds

of hospitalisation for women than men in Andhra Pradesh (odds ratio approximately 1.67, $p < 0.001$), adjusting for age and expenditure. Thus more focus on materal and women health care should be provided.

6.2 For Jammu And Kashmir

- Focus on the state-specific gender difference: Our logistic regression finds a statistically significant State × Gender interaction ($p < 0.001$). The female-versus-male hospitalisation odds ratio is higher in J&K relative to the corresponding odds ratio in Andhra Pradesh (interaction OR approximately 1.43). This supports investigating why the gender gap differs between the states. It does not, by itself, prove that women in J&K have worse health or face greater barriers.

In both states, policies to improve chronic disease prevention and management should be developed. Expand screening for diabetes, hypertension, and other chronic conditions through primary health centres and community health programmes. Since recorded chronic illness may rise with expenditure because wealthier people have better access to diagnosis, outreach should particularly reach lower-expenditure households.

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R-Code: https://github.com/Anushruta06/Demography_Project/R-code.r